SpeleoMesh

Using the SpeleoMesh

Dane Evans,
NSW Cave Rescue

Hardware

Types :

Trunk:

- Basic Internal antenna
- Basic External antenna
- Wired Link
- Solar nodes

Personal:

- GPS
- Various commercial designs

These the nodes that will make the majority of the trunk for the exercise. All are waterproof, with dual TPU gaskets.
The wired Nodes, and external antennas are more of a risk, so try not to dunk them, or place them in running water.

If you have any issues with any today, please let me know.

CAVING AUSTRALIA
Australian Cave Rescue Commission

VRA RESCUE NSW
ACQUIRO ET SUCCURRO

Hardware

Types :

Trunk:

- Basic Internal antenna
- Basic External antenna
- Wired Link
- Solar nodes

Personal:

- GPS
- Various commercial designs

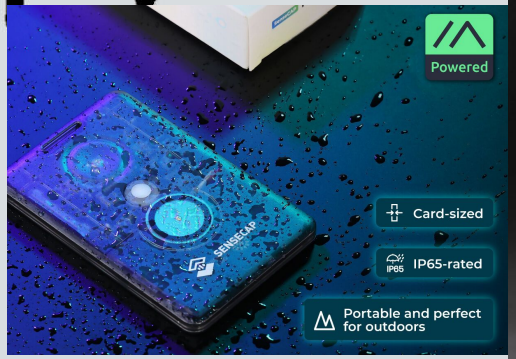
The GPS design hasn't been tested for water proofing - the gasket design should be adequate for water resistance, but if you are have one and are going anywhere near the water, please take care to place it in a ziplock bag or similar.
They have a much better bluetooth antenna, which is why we are using them as personal Nodes

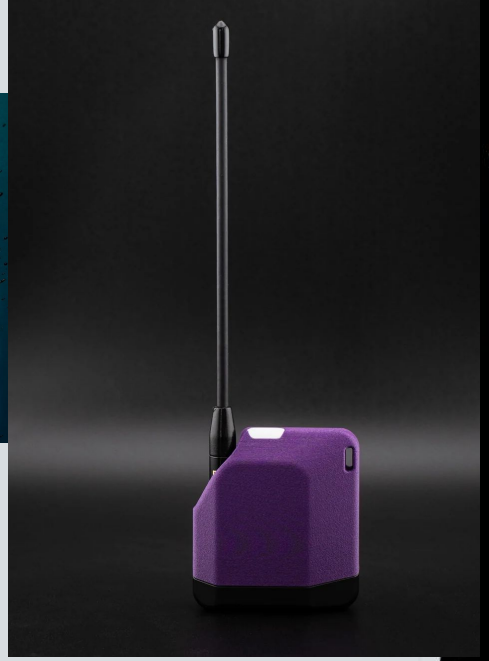
If you have any issues with any today, please let me know.

CAVING AUSTRALIA
Australian Cave Rescue Commission

VRA RESCUE NSW
ACQUIRO ET SUCCURRO

Hardware





Types :

Trunk:

- Basic Internal antenna
- Basic External antenna
- Wired Link
- Solar nodes

Personal:

- GPS
- Various commercial designs

There are several Commercial designs floating around - the card style T1000E are waterproof, as are the R1 Neos (theoretically), all of the others must be treated as if they aren't.

If you have any issues with any today, please let me know.

CAVING AUSTRALIA

Australian Cave Rescue Commission

VRA RESCUE NSW ACQUIRO ET SUCCURRO

Getting the app

MESHTASTIC

App links :

Droid -
Play store -
Meshtastic

Apple -
App store -
Meshtastic

CAVING AUSTRALIA
Australian Cave Rescue Commission

Setup sheet :

Network Connection

It's useful to know if the Trunk can still hear from the surface - It doesn't guarantee symmetrical communications.

All of the CaveNodes have a dedicated LED, similar to the Rangetest one, which can respond to messages from the surface.

- This defaults to RED,
- and is set to Green on a schedule from the surface.
- If you ever see RED, these messages are not getting through.
    - This could be because the server has crashed, or been turned off
    - It could be a single dropped message,
    - Or an actual problem.
- Try messaging the surface, see if there is a response.
- Ask teams above you what they can see / receive.
- If your unit appears to be the break, then send messages saying that it is broken, and where.

CAVING AUSTRALIA

Australian Cave Rescue Commission

VRA RESCUE NSW

ACQUIRO ET SUCCURRO

Placing Nodes

- Think like you do in a survey - maximise line of sight both forwards & back.
    - Place high where possible & suitable.
    - Human bodies block signal - try not to route through gaps where people will be spending a lot of time. We had an issue in the NSW exercise where I believe that was the cause.

- Rangetest LED and function
    - One of the additional LEDs in the device will respond to rangetest packets,
    - Green = good, yellow = ok, and red = marginal.
    - direct message to the placed node :
        - ADRT ON / ADRT Delay 15|30|60 / ADRT OFF
        - look for return messages in infra or the light to turn green.
            - Check for reliable reception (SNR > 3, RSSI > -100), & getting every packet

CAVING AUSTRALIA

Australian Cave Rescue Commission

VRA RESCUE NSW

ACQUIRO ET SUCCURRO

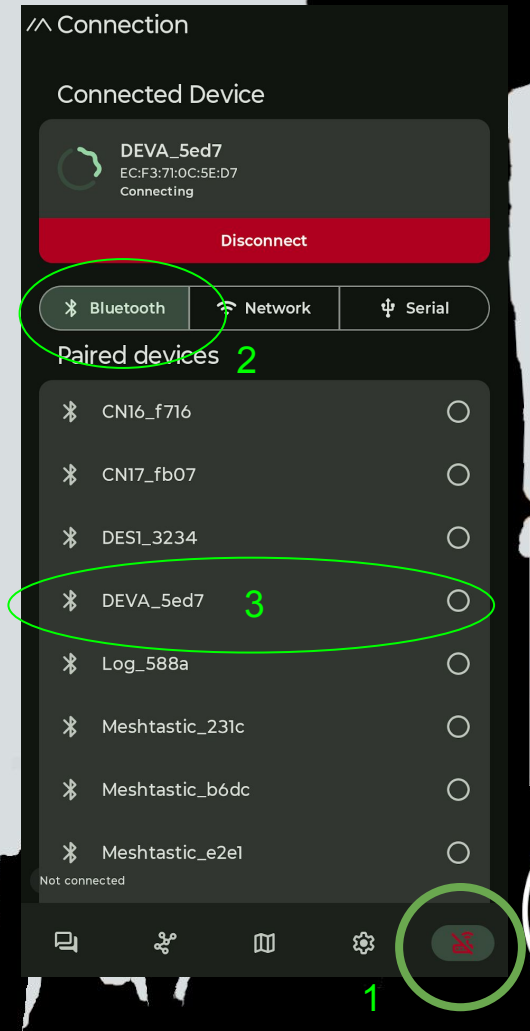
Connecting

Instructions are for Android - When I last tried Apple was similar, but a slightly different layout.

1. Connections tab
2. Select Bluetooth (Pin code is 123456)
3. Choose the node that matches the one in your hand - double check the last 4 digits
   a. Apple Doesn't use the node name here, just the initial.

- **Connect only to the node in your hand.**
- **Do not connect multiple phones to a single node - it doesn't work.**

And please forget the node in your bluetooth settings afterwards, unless you have been allocated it as a personal node for the exercise.



CAVING AUSTRALIA

Australian Cave Rescue Commission

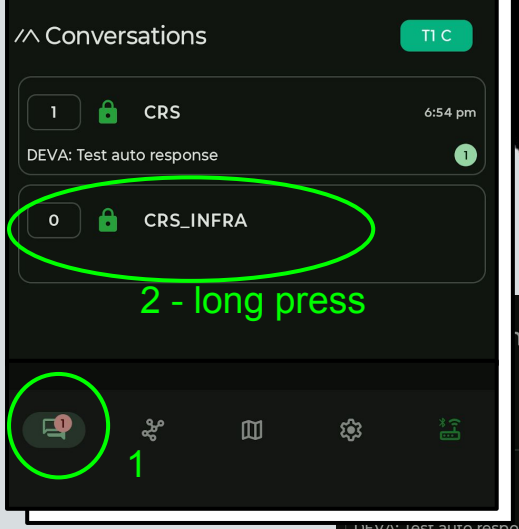
VRA RESCUE NSW
ACQUIRO ET SUCCURRO

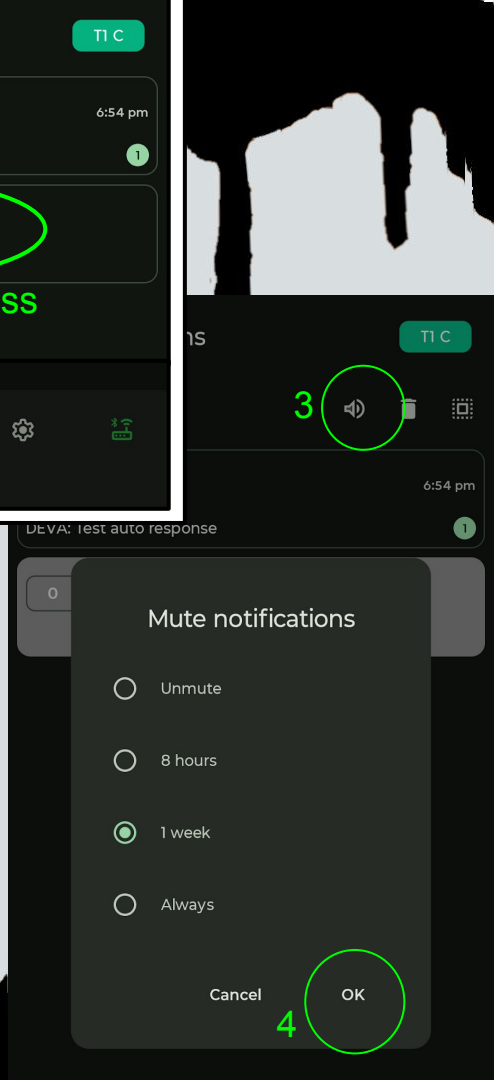
Channels

If you go to the channels Tab, you should see 2 :
0 - CRS INFRA
1 - CRS
There may be additional ones if that node has been involved in direct messages in the past

Mute the INFRAstructure channel - it's where all of the rangetest packets, position and other telemetry packets are sent, and for most people it will be pure noise.

- All channels and messages use the same bandwidth, keep traffic minimal where possible.
- Messages will take some minutes to traverse the network, do not resend quickly, or jump on immediately after someone else.





Practical exercises

1) Open the connections tab, Bluetooth 123456 - one at a time. if you don't have one allocated to you for Saturday, please 'forget' it afterwards, as it will autopair, and they don't handle conflicts well.
2) Get familiar with the other pages :
   - Messages,
   - Nodes
   - *Settings* 💀 **There is nothing in here for you, and a lot of settings that could kick you off the network.** ***If you don't know what you touched in here, let me know and I'll reset it.***

Please make sure you can :
- Send a message
- Recieve messages
- Observe the LEDs through RTs
   - Comms LED

CAVING AUSTRALIA
Australian Cave Rescue Commission

VRA RESCUE NSW
ACQUIRO ET SUCCURRO

Prac wrap up

If you have a personal node for the exercise:

- Note the number if you're handing it back to me.
- Make sure it's charged and you bring it
- If you have a t1000E, you can get audio notifications,
- If you have a smart watch, notifications on that may be possible.

If you don't have a personal node:

- Please delete the node from your bluetooth connections in your phone OS settings.
    - They auto pair, and don't handle conflicts well.

CAVING AUSTRALIA

Australian Cave Rescue Commission

VRA RESCUE NSW

ACQUIRO ET SUCCURRO

In the Major exercise

- Keep an eye on your node and / or app, you can set notifications in the App.
    - Being asynchronous, this need only be occasionally, or when you hear a notification
- Keep an eye on any nearby Nodes, look for red vs green lights, If it goes red, and you still have comms to the surface, send a message logging that the Comms LED is red.
- If you lose comms to the surface, post that message and ask the team above to check etc - the more info you can give me about the location, the faster I can get it patched.

CAVING AUSTRALIA

Australian Cave Rescue Commission

VRA RESCUE NSW

ACQUIRO ET SUCCURRO